BEE 4750 Homework 5: Mixed Integer and Stochastic Programming

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Due Date

Thursday, 12/04/24, 9:00pm

Overview

Instructions

- In Problem 1, you will use mixed integer programming to solve a waste load allocation problem.
- In Problem 2, you will formulate a stochastic optimization problem.

Load Environment

The following code loads the environment and makes sure all needed packages are installed. This should be at the start of most Julia scripts.

```
import Pkg
Pkg.activate(@__DIR__)
Pkg.instantiate()
```

```
  Activating project at `~/Documents/School/Senior/BEE 4750/hw5-rha-sims_hw5`
   Installed GR_jll         v0.73.18+0
   Installed OpenBLAS32_jll     v0.3.29+0
   Installed HiGHS_jll       v1.12.0+0
```

```
  Installed PlotUtils       v1.4.4
  Installed Measures        v0.3.3
  Installed MutableArithmetics  v1.6.7
  Installed OpenSSL         v1.6.0
  Installed FFMPEG          v0.4.5
  Installed Pango_jll       v1.57.0+0
  Installed StaticArraysCore    v1.4.4
  Installed JSON            v1.3.0
  Installed DataStructures     v0.19.3
  Installed Adapt           v4.4.0
  Installed GraphRecipes      v0.5.15
  Installed StatsBase       v0.34.8
  Installed StableRNGs       v1.0.4
  Installed HiGHS           v1.20.1
  Installed METIS_jll       v5.1.3+0
  Installed ForwardDiff      v1.3.0
  Installed StructUtils      v2.6.0
  Installed FFMPEG_jll       v8.0.0+0
  Installed JuMP            v1.29.3
  Installed GR             v0.73.18
Precompiling project...
  1139.3 ms   StableRNGs
  1142.9 ms   StaticArraysCore
  1407.0 ms   Measures
  1163.3 ms   Adapt
  1050.1 ms   METIS_jll
  1168.2 ms   StructUtils
  1907.8 ms   ChainRulesCore
   918.6 ms   OpenBLAS32_jll
  3443.1 ms   DataStructures
  1435.9 ms   ColorVectorSpace → SpecialFunctionsExt
  2855.6 ms   OpenSSL
  2584.7 ms   Pango_jll
   931.6 ms   DiffResults
  2607.9 ms   FFMPEG_jll
   765.6 ms   OffsetArrays → OffsetArraysAdaptExt
  1268.3 ms   Adapt → AdaptSparseArraysExt
  1242.9 ms   StructUtils → StructUtilsTablesExt
  8264.9 ms   MutableArithmetics
  7339.7 ms   ColorSchemes
  1253.7 ms   ChainRulesCore → ChainRulesCoreSparseArraysExt
  1766.4 ms   LogExpFunctions → LogExpFunctionsChainRulesCoreExt
  2222.6 ms   SpecialFunctions → SpecialFunctionsChainRulesCoreExt
```

```
  1165.7 ms    SortingAlgorithms
  5348.8 ms    JSON
  4218.1 ms    HiGHS_jll
 10813.3 ms    StaticArrays
  4109.5 ms    FFMPEG
  6367.0 ms    ForwardDiff
  5200.7 ms    GR_jll
  3886.0 ms    StatsBase
  2279.4 ms    BenchmarkTools
  1146.7 ms    StaticArrays → StaticArraysChainRulesCoreExt
  2354.3 ms    ArnoldiMethod
  1050.7 ms    StaticArrays → StaticArraysStatisticsExt
  1031.2 ms    Adapt → AdaptStaticArraysExt
  1990.2 ms    ForwardDiff → ForwardDiffStaticArraysExt
 16360.0 ms    PlotUtils
  3571.0 ms    GeometryTypes
 29107.5 ms    HTTP
 18869.8 ms    GeometryBasics
  7418.8 ms    Graphs
  4755.4 ms    Interpolations
  4585.2 ms    PlotThemes
  3238.5 ms    NetworkLayout
  6590.9 ms    RecipesPipeline
  2408.0 ms    Interpolations → InterpolationsForwardDiffExt
  7349.6 ms    GR
  3113.5 ms    NetworkLayout → NetworkLayoutGraphsExt
  4677.3 ms    GraphRecipes
 73881.4 ms    DataFrames
 66668.3 ms    MathOptInterface
  3844.2 ms    Latexify → DataFramesExt
  3249.5 ms    MathOptIIS
 13480.3 ms    HiGHS
 24224.9 ms    JuMP
 79723.2 ms    Plots
  4658.6 ms    Plots → GeometryBasicsExt
 57 dependencies successfully precompiled in 139 seconds. 195 already precompiled.
```

```julia
using JuMP
using HiGHS
using DataFrames
using GraphRecipes
using Plots
```

```
using Measures
using MarkdownTables
```

Problems (Total: 30 Points)

Problem 1 (24 points)

Three cities are developing a coordinated municipal solid waste (MSW) disposal plan. Three disposal alternatives are being considered: a landfill (LF), a materials recycling facility (MRF), and a waste-to-energy facility (WTE). The capacities of these facilities and the fees for operation and disposal are provided below.

- **LF**: Capacity 200 Mg, fixed cost \$2000/day, tipping cost \$50/Mg;
- **MRF**: Capacity 350 Mg, fixed cost \$1500/day, tipping cost \$7/Mg, recycling cost \$40/Mg recycled;
- **WTE**: Capacity 210 Mg, fixed cost \$2500/day, tipping cost \$60/Mg;

Transportation costs are \$1.5/Mg-km, and the relative distances between the cities and facilities are provided in the table below.

City/Facility	Landfill (km)	MRF (km)	WTE (km)
1	5	30	15
2	15	25	10
3	13	45	20
LF	-	32	18
MRF	32	-	15
WTE	18	15	-

The fixed costs associated with the disposal options are incurred only if the particular disposal option is implemented. The three cities produce 100, 90, and 120 Mg/day of solid waste, respectively, with the composition provided in the table below.

Component	% of total mass	Combustion ash (%)	MRF Recycling rate (%)
Food Wastes	15	8	0
Paper & Cardboard	40	7	55
Plastics	5	5	15
Textiles	3	10	10
Rubber, Leather	2	15	0
Wood	5	2	30

Component	% of total mass	Combustion ash (%)	MRF Recycling rate (%)
Yard Wastes	18	2	40
Glass	4	100	60
Ferrous	2	100	75
Aluminum	2	100	80
Other Metal	1	100	50
Miscellaneous	3	70	0

The information in the above table will help you determine the overall recycling and ash fractions. Note that the recycling residuals, which may be sent to either landfill or the WTE, have different ash content than the ash content of the original MSW. You will need to determine these fractions to construct your mass balance constraints.

Reminder: Use `round(x; digits=n)` to report values to the appropriate precision!

Problem 1.1

Based on the information above, calculate the overall recycling and ash fractions for the waste produced by each city.

> Overall recycle fraction: 37.75%
>
> Overall ash fraction: 16.41%
>
> These were calculated by taking the dot product of the % of total mass vector and the combustion ash % or MRF recycle rate % vectors to get the weighted average of the overall percentage for each.

Problem 1.2

What are the decision variables for your optimization problem? Provide notation and variable meaning.

The decision variables for are follows:

$W_{i,j}$ –> The amount of waste to send from each city (i) to each facility (j)

$R_{k,j}$ –> The amount of residual ash to send from each facility (k) to another facility (j) for disposal

Y_j –> Binary variable for operational status (on/off)

For indexing, cities are labeled 1-3, and facilities are LF = 4, MRF = 5, and WTE = 6 for i = [1,2,3], k = [5,6], j = [4,5,6]

```julia
waste = Model(HiGHS.Optimizer)
I = [1:3]
K = [5:6]
J = [4:6]
@variable(waste, W[i in I, j in J], Int)
@variable(waste, R[k in K, j in J], Int)
@variable(waste, Y[j in J], Bin)
```

```
1-dimensional DenseAxisArray{VariableRef,1,...} with index sets:
    Dimension 1, UnitRange{Int64}[4:6]
And data, a 1-element Vector{VariableRef}:
 Y[4:6]
```

Problem 1.3

Formulate the objective function. Make sure to include any needed derivations or justifications for your equation(s).

FC_j is fixed costs per facility j multiplied by the binary operational variable Y_j: $>FC_j * Y_j =$ fixed cost portion of objective

DC_j is dumping (tipping) costs per facility j multiplied by the amount of waste going to facility j which is sum of i across $W_{i,j}$ and k across $R_{k,j}$ with k not equal to j: $>DC_j * (\sum_i W_{i,j} + \sum_k R_{k,j}) =$ variable dumping cost portion of objective

RC_j is the recycle cost for facility j (only $j = 2$ for MRF will be nonzero) multiplied by r_j the recycle rate for facility j and multiplied again by the amount of waste going to facility j which is sum of i across $W_{i,j}$: $>RC_j * r_j * \sum_i W_{i,j} =$ variable recycle cost portion of objective

TC_i, j is transportation costs per Mg-km from facility/city i to facility j multiplied by the amount of waste going from facility/city i to j which is sum of i across $W_{i,j}$ and multiplied by the distance $d_{i,j}$ from facility/city i to facility j. Do the same for $R_{k,j}$ across k add that as well (with k not equal to j): $>TC_j * (\sum_i (d_{i,j} * W_{i,j}) + \sum_k (d_{k,j} * R_{k,j})) =$ variable transportation cost portion of objective

OVERALL OBJECTIVE FUNCTION:

$ min_{Y,W,R} _j [(FC_j * Y_j) + (DC_j * (*i W*{i,j} + *k R*{k,j})) + (RC_j * r_j * *i W*{i,j}) + (TC_j * (*i (d*{i,j} * W_{i,j}) + *k (d*{k,j} * R_{k,j}))))] $

$$min_{Y,W,R}[(FC_4*Y_4)+(FC_5*Y_5)+(FC_6*Y_6)+(DC_4*(\sum_i W_{i,4}+R_{5,4}+R_{6,4}))+(DC_5*(\sum_i W_{i,5}+R_{6,5}))+(DC_6*(\sum_i W_{i,6}+R_{5,6}))+(RC_5*r_5*\sum_i W_{i,5})+(TC_4*(\sum_i(d_{i,4}*W_{i,4})+(d_{5,4}*R_{5,4})+(d_{6,4}*R_{6,4})))+(TC_5*(\sum_i(d_{i,5}*W_{i,5})+(d_{6,5}*R_{6,5})))+(TC_6*(\sum_i(d_{i,6}*W_{i,6})+(d_{5,6}*R_{5,6})))]$$

```julia
waste = Model(HiGHS.Optimizer)
I = 1:3
K = 5:6
J = 4:6
@variable(waste, W[i in I, j in J], Int)
@variable(waste, R[k in K, j in J], Int)
@variable(waste, Y[j in J], Bin)

FC = [2000, 1500, 2500] # $/day for correspond to j=4,5,6
DC = [50, 7, 60] # $/Mg for j=4,5,6
RC = 40 # $/Mg for j=5 because its the only recycle facility
TC = 1.5 # $/Mg/km for all travel between each city and facility or between two facilities
r = 0.3775 #recycle rate
a = 0.1641 #ash production rate

#distances from facility/city i (rows) to j (columns)
d = [
    0 0 0 5 30 15
    0 0 0 15 25 10
    0 0 0 13 45 20
    0 0 0 0 32 18
    0 0 0 32 0 15
    0 0 0 18 15 0
]

@objective(waste, Min,
                    #fixed costs
                    FC[1]*Y[4] + FC[2]*Y[5] + FC[3]*Y[6]
                    #dumping (tipping costs)
                    + DC[1]*(sum(W[i,4] for i in I)+R[5,4]+R[6,4])
                    + DC[2]*(sum(W[i,5] for i in I)+R[6,5])
                    + DC[3]*(sum(W[i,6] for i in I)+R[5,6])
                    #recycle costs
                    + RC*r*sum(W[i,5] for i in I)
                    #transportation costs
                    + TC*(sum(d[i,4]*W[i,4] for i in I) + d[5,4]*R[5,4] + d[6,4]*R[6,4])
                    + TC*(sum(d[i,5]*W[i,5] for i in I) + d[6,5]*R[6,5])
                    + TC*(sum(d[i,6]*W[i,6] for i in I) + d[5,6]*R[5,6])
)
```

$2000Y_4 + 1500Y_5 + 2500Y_6 + 57.5W_{1,4} + 72.5W_{2,4} + 69.5W_{3,4} + 98R_{5,4} + 77R_{6,4} + 67.1W_{1,5} + 59.6W_{2,5} + 89.6W_{3,5} + 29.5R_{6,5} + 82.5W_{1,6} + 75W_{2,6} + 90W_{3,6} + 82.5R_{5,6}$

Problem 1.4

Derive all relevant constraints. Make sure to include any needed justifications or derivations.

Problem 1.5

Find the optimal solution (using `JuMP` to solve the problem). Report the optimal objective value.

Problem 1.6

Draw a diagram showing the flows of waste between the cities and the facilities. Which facilities (if any) will not be used? Does this solution make sense?

Problem 2 (6 points)

Consider a two-period economic dispatch problem, based on the multi-period example from Lecture 14 (on 10/29). The generator data, including ramping constraints for each generator, is provided in 'data/generators.csv.' In period 1, the demand is $d_1 = 1100\text{MW}$. In period 2, the demand is projected to be $d_2 = 1200\text{MW}$, but there is a 25% probability that it is \$1500 . In the first period, the solar capacity factor is 0.9 and the wind capacity factor is 0.45, but in the second period, there is some uncertainty: the forecasted solar and wind capacity factors are 0.95 and 0.4, respectively, but there is a 30% probability that they are 0.75 and 0.5. Your goal is to identify how to dispatch your generators to minimize the cost of meeting demand.

Problem 2.1

Draw a scenario tree for this problem.

Problem 2.2

Formulate a stochastic linear program for this problem based on your scenario tree from Problem 2.1 and the data in `data/generators.csv`.

References

List any external references consulted, including classmates.